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WIRELESS SENSING FOR TRANSMISSIVE AND HYBRID RECONFIGURABLE INTELLIGENT SURFACES

Abstract

A wireless device may transmit, for a network node, a capability report including (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The wireless device may receive, from the network node, a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The wireless device may select, based on the transmissive signal configuration or the transmissive and reflective configuration: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to communication systems, and more particularly, to a wireless sensing system using a reconfigurable intelligent surface (RIS).

INTRODUCTION

[0002] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

[0003] These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

[0004] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0005] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus at a wireless device are provided. The wireless device may be a reconfigurable intelligent surface (RIS). The apparatus may transmit, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The apparatus may receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The apparatus may select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission

direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

[0006] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus at a network node are provided. The apparatus may receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The apparatus may transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

[0007] To the accomplishment of the foregoing and related ends, the one or more aspects may include the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating an example of a wireless communications system and an access network.

[0009] FIG. 2A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.

[0010] FIG. 2B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.

[0011] FIG. 2C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.

[0012] FIG. 2D is a diagram illustrating an example of uplink (UL) channels within a subframe, in accordance with various aspects of the present disclosure.

[0013] FIG. 3 is a diagram illustrating an example of a base station and user equipment (UE) in an access network.

[0014] FIG. 4 is a diagram illustrating an example of sensing based on measurements of sensing signals reflected off of a target object, in accordance with various aspects of the present disclosure.

[0015] FIG. 5 is a diagram illustrating an example of a wireless device configured to reflect, transmit, or reflect and transmit one or more signals from a network node, in accordance with various aspects of the present disclosure.

[0016] FIG. 6A is a diagram illustrating an example of meta-elements of a wireless device configured to transmit a signal from a network node to a UE, in accordance with various aspects of the present disclosure.

[0017] FIG. 6B is a diagram illustrating another example of meta-elements of a wireless device configured to transmit a signal from a network node to a UE, in accordance with various aspects of the present disclosure.

[0018] FIG. 7A is a diagram illustrating an example of a wireless device configured to assist a network node in performing monostatic sensing of a target object, in accordance with various aspects of the present disclosure.

[0019] FIG. 7B is a diagram illustrating an example of a wireless device configured to assist a network node in performing bistatic sensing of a target object, in accordance with various aspects

of the present disclosure.

[0020] FIG. 7C is a diagram illustrating an example of a wireless device configured to assist a network node in performing monostatic or bistatic sensing of a target object using a plurality of differently configured surfaces, in accordance with various aspects of the present disclosure.

[0021] FIG. 8A is a diagram illustrating another example of a wireless device configured to assist a network node in performing monostatic sensing of a target object using a plurality of differently configured surfaces, in accordance with various aspects of the present disclosure.

[0022] FIG. 8B is a diagram illustrating another example of a wireless device configured to assist a network node in performing bistatic sensing of a target object using a plurality of differently configured surfaces, in accordance with various aspects of the present disclosure.

[0023] FIG. 9A is a diagram illustrating another example of a wireless device configured to assist a network node in performing monostatic sensing of a target object, in accordance with various aspects of the present disclosure.

[0024] FIG. 9B is a diagram illustrating another example of a wireless device configured to assist a network node in performing monostatic sensing of a target object, in accordance with various aspects of the present disclosure.

[0025] FIG. 10A is a diagram illustrating an example of a wireless device configured to assist a network node in performing monostatic sensing of a plurality of target objects using both transmissive and reflective properties of its meta-elements, in accordance with various aspects of the present disclosure.

[0026] FIG. 10B is a diagram illustrating an example of a wireless device configured to assist a network node in performing monostatic sensing of a plurality of target objects using both transmissive and reflective properties of its meta-elements with a plurality of differently configured surfaces, in accordance with various aspects of the present disclosure.

[0027] FIG. 11 is a connection flow diagram illustrating an example of communications between a network node and a wireless device configured to assist a network node in performing sensing on a target object, in accordance with various aspects of the present disclosure.

[0028] FIG. 12 is a flowchart of a method of wireless communication.

[0029] FIG. 13 is another flowchart of a method of wireless communication.

[0030] FIG. 14 is a diagram illustrating an example of a hardware implementation for an example network entity.

[0031] FIG. 15 is a diagram illustrating an example of a hardware implementation for an example network entity.

DETAILED DESCRIPTION

[0032] The detailed description set forth below in connection with the drawings describes various configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0033] Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0034] By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units

(GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

[0035] Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media may include a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

[0036] While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

[0037] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmission reception point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

[0038] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0039] Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0040] FIG. 1 is a diagram **100** illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs **110** that can communicate directly with a core network **120** via a backhaul link, or indirectly with the core network **120** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **125** via an E2 link, or a Non-Real Time (Non-RT) RIC **115** associated with a Service Management and Orchestration (SMO) Framework **105**, or both). A CU **110** may communicate with one or more DUs **130** via respective midhaul links, such as an F1 interface. The DUs **130** may communicate with one or more RUs **140** via respective fronthaul links. The RUs **140** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **140**.

[0041] Each of the units, i.e., the CUS **110**, the DUs **130**, the RUs **140**, as well as the Near-RT RICs **125**, the Non-RT RICs **115**, and the SMO Framework **105**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0042] In some aspects, the CU **110** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **110**. The CU **110** may be configured to handle user plane functionality (i.e., Central Unit—User Plane (CU-UP)), control plane functionality (i.e., Central Unit—Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate

bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

[0043] The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **140**. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

[0044] Lower-layer functionality can be implemented by one or more RUs **140**. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **140** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **140** can be controlled by the corresponding DU **130**. In some scenarios, this configuration can enable the DU(s) **130** and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0045] The SMO Framework **105** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **110**, DUs **130**, RUs **140** and Near-RT RICs **125**. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs **140** via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to support functionality of the SMO Framework **105**.

[0046] The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **110**, one or more DUs **130**, or both, as well as an O-eNB, with the Near-RT RIC **125**.

[0047] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network

functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

[0048] At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base stations **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs **140** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

[0049] Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

[0050] The wireless communications system may further include a Wi-Fi AP **150** in communication with UEs **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UEs **104**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

[0051] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

[0052] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies.

Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FRI characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0053] With the above aspects in mind, unless specifically stated otherwise, the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

[0054] The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **102**/UE **104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

[0055] The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmission reception point (TRP), network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

[0056] The core network **120** may include an Access and Mobility Management Function (AMF) **161**, a Session Management Function (SMF) **162**, a User Plane Function (UPF) **163**, a Unified Data Management (UDM) **164**, one or more location servers **168**, and other functional entities. The AMF **161** is the control node that processes the signaling between the UEs **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) **165** and a Location Management Function (LMF) **166**. However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC **165**, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to

compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the serving base station **102**. The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network (NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TBS), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

[0057] Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

[0058] A wireless device **106** may be a meta-surface configured to receive an incident wave from another wireless device, such as a base station **102**, an RU **140** of a base station **102**, or a UE **104**. The wireless device **106** may be a reconfigurable intelligent surface (RIS). The wireless device **106** may be configured to redirect a transmission through the wireless device **106** (e.g., generate a transmissive signal from a signal received by the wireless device **106**) towards a desired direction or reflect a transmission off of a surface of the wireless device **106** (e.g., generate a reflective signal from a signal received by the wireless device **106**), towards a desired direction, or redirect a portion of the incident wave and reflect another portion of the incident wave.

[0059] Referring again to FIG. **1**, in certain aspects, the wireless device **106** may have a penetrative signal component **198** configured to transmit, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The penetrative signal component **198** may be configured to receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The penetrative signal component **198** may be configured to select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. In certain aspects, the base station **102** may have a penetrative element configuration component **199** configured to receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3)

a third indication of a transmissive and reflective capability. The penetrative element configuration component **199** may be configured to transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. Although the following description may be focused on RIS devices, the concepts described herein may be applicable to any wireless device that may be configured to redirect and/or reflect signals off of a surface of the device based on a configuration received from another wireless device. Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and other wireless technologies.

[0060] FIG. 2A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. 2B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. 2C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. 2D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. 2A, 2C, the 5G NR frame structure is assumed to be TDD, with subframe **4** being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe **3** being configured with slot format 1 (with all UL). While subframes **3**, **4** are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

[0061] FIGS. 2A-2D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (also referred to as single carrier frequency-division multiple access (SC-FDMA) symbols) (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with $1/\text{SCS}$.

TABLE-US-00001 TABLE 1 Numerology, SCS, and CP SCS μ $\Delta f = 2^{\mu} \cdot 15$ [kHz]
Cyclic prefix 0 15 Normal 1 30 Normal 2 60 Normal, Extended 3 120 Normal 4 240 Normal 5 480 Normal 6 960 Normal

[0062] For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per

subframe. Accordingly, for normal CP and numerology μ , there are 14 symbols/slot and 24 slots/subframe. The subcarrier spacing may be equal to $2^{\mu} \cdot 15$ kHz, where μ is the numerology 0 to 4. As such, the numerology $\mu=0$ has a subcarrier spacing of 15 kHz and the numerology $\mu=4$ has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. 2A-2D provide an example of normal CP with 14 symbols per slot and numerology $\mu=2$ with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μ s. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. 2B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

[0063] A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

[0064] As illustrated in FIG. 2A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).

[0065] FIG. 2B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE 104 to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.

[0066] As illustrated in FIG. 2C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

[0067] FIG. 2D illustrates an example of various UL channels within a subframe of a frame. The

PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

[0068] FIG. 3 is a block diagram of a base station 310 in communication with a UE 350 in an access network. In the DL, Internet protocol (IP) packets may be provided to a controller/processor 375. The controller/processor 375 implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor 375 provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0069] The transmit (Tx) processor 316 and the receive (Rx) processor 370 implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The Tx processor 316 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator 374 may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 350. Each spatial stream may then be provided to a different antenna 320 via a separate transmitter 318Tx. Each transmitter 318Tx may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.

[0070] At the UE 350, each receiver 354Rx receives a signal through its respective antenna 352. Each receiver 354Rx recovers information modulated onto an RF carrier and provides the information to the receive (Rx) processor 356. The Tx processor 368 and the Rx processor 356 implement layer 1 functionality associated with various signal processing functions. The Rx processor 356 may perform spatial processing on the information to recover any spatial streams destined for the UE 350. If multiple spatial streams are destined for the UE 350, they may be combined by the Rx processor 356 into a single OFDM symbol stream. The Rx processor 356 then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast

Fourier Transform (FFT). The frequency domain signal may include a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements layer 3 and layer 2 functionality.

[0071] The controller/processor **359** can be associated with a memory **360** that stores program codes and data. The memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0072] Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0073] Channel estimates derived by a channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the Tx processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the Tx processor **368** may be provided to different antenna **352** via separate transmitters **354Tx**. Each transmitter **354Tx** may modulate an RF carrier with a respective spatial stream for transmission.

[0074] The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318Rx** receives a signal through its respective antenna **320**. Each receiver **318Rx** recovers information modulated onto an RF carrier and provides the information to a Rx processor **370**.

[0075] The controller/processor **375** can be associated with a memory **376** that stores program codes and data. The memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0076] At least one of the Tx processor **316**, the Rx processor **370**, and the controller/processor **375** may be configured to perform aspects in connection with the penetrative element configuration component **199** of FIG. 1.

[0077] FIG. 4 is a diagram **400** illustrating an example of sensing based on sensing signal measurements. In one aspect, the wireless node **402** may perform monostatic sensing, where the wireless node **402** may transmit a set of sensing signals **412** at the target object **403**, the target object **403** may reflect the set of sensing signals **412** as the reflected set of sensing signals **416** at the wireless node **402**, and the wireless node **402** may measure the reflected set of sensing signals **416** from the target object **403**. In another aspect, the wireless node **402** and the wireless node **404**

may perform bistatic sensing, where the wireless node **402** may transmit a set of sensing signals **412** at the target object **403**, the target object **403** may reflect the set of sensing signals **412** as the reflected set of sensing signals **414** at the wireless node **404**, and the wireless node **404** may measure the reflected set of sensing signals **414** from the target object **403**. In another aspect the wireless node **402** and the wireless node **406** may perform multi-static sensing, where in addition to the wireless node **402** measuring the reflected set of sensing signals **416** from the target object **403** using monostatic sensing, the wireless node **406** may transmit a set of sensing signals **418** at the target object **403**, the target object **403** may reflect the set of sensing signals **418** as the reflected set of sensing signals **420** at the wireless node **402**, and the wireless node **402** may measure the reflected set of sensing signals **420** from the target object **403**. In another aspect the wireless node **402**, the wireless node **404**, and the wireless node **408** may perform multi-static sensing, where in addition to the wireless node **404** measuring the reflected set of sensing signals **414** from the target object **403** using bistatic sensing, the wireless node **408** may transmit a set of sensing signals **422** at the target object **403**, the target object **403** may reflect the set of sensing signals **422** as the reflected set of sensing signals **424** at the wireless node **404**, and the wireless node **404** may measure the reflected set of sensing signals **424** from the target object **403**. Each wireless node may be any wireless device configured to transmit or receive wireless signals, such as UEs, network nodes, TRPs, or base stations. For example, the wireless node **402** may be a network node configured to transmit the set of sensing signals **412** at the target object **403** and measure the reflected set of sensing signals **416** from the target object **403**. In another example, the wireless node **402** may be a network node configured to transmit the set of sensing signals **412** at the target object **403**, and the wireless node **404** may be a UE configured to measure the reflected set of sensing signals **414** from the target object **403**.

[0078] The wireless node **402** may conduct one or more sensing measurements on the reflected set of sensing signals **416** and/or the reflected set of sensing signals **420**. In one aspect, the wireless node **402** may calculate a distance or a range between the wireless node **402** and the target object **403** based on a round trip time (RTT) between when the wireless node **402** transmits the set of sensing signals **412** and when the wireless node **402** receives the reflected set of sensing signals **416**. In one aspect, the wireless node **402** may calculate a distance or a range that the set of sensing signals **418** and the reflected set of sensing signals **420** travels based on a time between when the wireless node **406** transmits the set of sensing signals **418** and when the wireless node **402** receives the reflected set of sensing signals **420**. In one aspect, the wireless node **402** may calculate a location of the target object **403** based on a plurality or range or distance measurements, for example via triangulation using known positions of the wireless nodes **402** and **406** and the calculated range or distance measurements. In one aspect, the wireless node **402** may calculate a velocity of the target object **403** based on a first calculated location of the target object **403** based on the reflected set of sensing signals **416** and/or the reflected set of sensing signals **420** measured at a first time, and a second calculated location of the target object **403** based on the reflected set of sensing signals **416** and/or the reflected set of sensing signals **420** measured at a second time. In one aspect, the wireless node **402** may calculate an AoA of the reflected set of sensing signals **416** and/or an AoD of the set of sensing signals **412** based on a plurality of ports that transmitted the set of sensing signals **412** and a plurality of ports that received the reflected set of sensing signals **416**. In one aspect, the wireless node **402** may calculate an AoA of the reflected set of sensing signals **420** and/or an AoD of the set of sensing signals **418** based on a plurality of ports that transmitted the set of sensing signals **418** and a plurality of ports that received the reflected set of sensing signals **420**.

[0079] Similarly, the wireless node **404** may conduct one or more sensing measurements on the reflected set of sensing signals **414** and/or the reflected set of sensing signals **424**. In one aspect, the wireless node **404** may calculate a distance or a range that the set of sensing signals **412** and the reflected set of sensing signals **414** travels based on a on a time between when the wireless node

402 transmits the set of sensing signals **412** and when the wireless node **404** receives the reflected set of sensing signals **414**. In one aspect, the wireless node **404** may calculate a distance or a range that the set of sensing signals **422** and the reflected set of sensing signals **424** travels based on a time between when the wireless node **408** transmits the set of sensing signals **422** and when the wireless node **404** receives the reflected set of sensing signals **424**. In one aspect, the wireless node **404** may calculate a location of the target object **403** based on a plurality or range or distance measurements, for example via triangulation using the known positions of wireless nodes **402**, **404**, and **408**, and the calculated range or distance measurements. In one aspect, the wireless node **404** may calculate a velocity of the target object **403** based on a first calculated location of the target object **403** based on the reflected set of sensing signals **414** and/or the reflected set of sensing signals **424** measured at a first time, and a second calculated location of the target object **403** based on the reflected set of sensing signals **414** and/or the reflected set of sensing signals **424** measured at a second time. In one aspect, the wireless node **404** may calculate an AoA of the reflected set of sensing signals **414** and/or an AoD of the set of sensing signals **412** based on a plurality of ports that transmitted the set of sensing signals **412** and a plurality of ports that received the reflected set of sensing signals **414**. In one aspect, the wireless node **404** may calculate an AoA of the reflected set of sensing signals **424** and/or an AoD of the set of sensing signals **422** based on a plurality of ports that transmitted the set of sensing signals **422** and a plurality of ports that received the reflected set of sensing signals **424**.

[0080] A network node or a UE configured to perform measurements on a set of reflected sensing signals may be configured to transmit a sensing signal report to a sensing server (e.g., an LMF) that coordinates a plurality of wireless nodes to perform sensing on a target object. In order to perform Doppler estimates or velocity estimates of a target object, such as the target object **403** in FIG. 4, or of a UE, such as the UE **104** in FIG. 1, the receiver wireless node may be configured to measure a reflected set of sensing signals at multiple points of time.

[0081] FIG. 5 is a diagram **500** illustrating an example of a wireless device **506** configured to receive a signal **503** from a network node **502**. The wireless device **506** may be, for example, a window of an enclosed space **510** (e.g., a building, a wall, or a vehicle) having walls or panels that block wireless signals, but the wireless device **506** may redirect and/or reflect wireless signals. Using the meta-elements **507** at the surface of the wireless device **506**, the wireless device **506** may redirect the signal **503** as the transmissive signal **509** towards the UE **508**. The UE **508** may then measure the transmissive signal **509** to perform sensing on the wireless device **506**. In some aspects, the wireless device **506** may redirect the transmissive signal **509** towards a target object, which then may reflect the transmissive signal to a wireless device, such as a network node or a UE, for sensing the target object. Using the meta-elements **507** at the surface of the wireless device **506**, the wireless device **506** may reflect the signal **503** as the reflective signal **505** towards the UE **504**. The UE **504** may then measure the reflective signal **505** to perform sensing on the wireless device **506**. In some aspects, the wireless device **506** may reflect the reflective signal **505** towards a target object, which then may reflect the reflective signal to a wireless device, such as a network node or a UE, for sensing the target object. One or more of the meta-elements **507** of a meta-surface of the wireless device **506** may have a transmissive mode, a reflective mode, or a hybrid transmissive and reflective mode.

[0082] When a meta-element **507** is switched to a reflection mode, the meta-element **507** may be configured to reflect the signal **503** to a desired direction. In other words, the wireless device **506** may reflect the signal **503** to an object on the same side of the network node **502**. The configuration of one or more reflective elements, such as a meta-element **507**, may be used to aim a signal **503** in a desired direction. For example, one or more reflection coefficients of the meta-element **507** may be changed to alter a direction that the reflective signal **505** is centered upon. For example, a first coefficient may be altered to change an amplitude of the reflective signal **505** from the meta-element **507** and a second coefficient may be altered to shift a phase of the reflective signal **505**

from the meta-element **507**. The configuration of the meta-element **507** of the wireless device **506** may depend on the knowledge of the direction of the incident wave of the signal **503**. In other words, the accuracy of where a meta-element **507** centers or aims the reflective signal **505** may be increased using information about the direction that the signal **503** approaches the meta-element **507** from, or an AoA of the signal **503** relative to the meta-element **507**. In some aspects, the network node **502** may transmit a configuration to the wireless device **506** to assist the wireless device **506** in accurately aiming the reflective signal **505**.

[0083] Similarly, when a meta-element **507** is switched to a transmissive mode, the meta-element **507** may be configured to redirect the signal **503** to a desired direction. In other words, the wireless device **506** may redirect the signal **503** to an object on the opposite side of the network node **502** (i.e., the signal **503** may penetrate the enclosed space **510** via the wireless device **506**). The configuration of one or more redirecting elements, such as a meta-element **507**, may be used to redirect a signal **503** towards a desired direction, similar to a prism. For example, one or more redirection coefficients of the meta-element **507** may be changed to alter a direction that the transmissive signal **509** is centered upon. For example, a first coefficient may be altered to change an amplitude of the transmissive signal **509** via the meta-element **507** and a second coefficient may be altered to shift a phase of the transmissive signal **509** via the meta-element **507**. The configuration of the meta-element **507** of the wireless device **506** may depend on the knowledge of the direction of the incident wave of the signal **503**. In other words, the accuracy of where a meta-element **507** centers or aims the transmissive signal **509** may be increased using information about the direction that the signal **503** approaches the meta-element **507** from, or an AoA of the signal **503** relative to the meta-element **507**. In some aspects, the network node **502** may transmit a configuration to the wireless device **506** to assist the wireless device **506** in accurately aiming the transmissive signal **509**.

[0084] In some aspects, the wireless device **506** may be configured to redirect a signal **503** in any direction relative to the meta-element **507** by either redirecting the signal **503** as a transmissive signal **509** or by reflecting the signal **503** as a reflective signal **505**.

[0085] In some aspects, the wireless device **506** may be configured to activate both a transmissive mode and a reflective mode. In other words, the wireless device **506** may have a dual function of reflection and transmission of the signal **503** to a first object on the same side of the network node **502** and to a second object on the opposite side of the network node **502**, respectively. For example, the wireless device **506** may be configured to reflect a portion of the signal **503** as the reflective signal **505** towards the UE **504**, and may be configured to redirect a portion of the signal **503** towards the UE **508**. In other words, one or more meta-elements **507** of the wireless device **506** may be configured to have both reflective and transmissive capabilities such that the meta-surface of the wireless device **506** may reflect one portion of an impinging signal in a controllable manner, while simultaneously redirecting another portion of the impinging signal in a controllable manner. The wireless device **506** may be described as having a hybrid transmissive and reflective mode. In some aspects, one or more meta-elements **507** may be switched to a transmissive mode while one or more other meta-elements may be switched to a reflective mode.

[0086] The network node **502** may have a penetrative element configuration component **199**. The penetrative element configuration component **199** may be configured to receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The penetrative element configuration component **199** may be configured to transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of

the surface of the wireless device for a reflective mode.

[0087] The wireless device **506** may have a penetrative signal component **198**. The penetrative signal component **198** may be configured to transmit, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The penetrative signal component **198** may be configured to receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The penetrative signal component **198** may be configured to select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

[0088] As discussed supra, the penetrative signal component **198** may be configured to transmit, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The penetrative signal component **198** may be configured to receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The penetrative signal component **198** may be configured to select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The penetrative signal component **198** may be within a processor of the wireless device **506**. The penetrative signal component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. In one configuration, the wireless device **506** may include means for transmitting, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The wireless device **506** may include means for receiving, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The wireless device **506** may include means for selecting, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The capability report may include a fourth indication of whether the first transmissive coefficient value is associated with monodirectional transmission or bidirectional transmission. The capability report may include a fourth indication of a bidirectional transmission capability. The first transmissive coefficient value may be associated with the first transmission direction. The capability report may further include a fourth indication of a second transmissive coefficient value associated with the second transmission direction. The first transmissive coefficient value may be different from the second transmissive coefficient value. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of the first section of the surface of the wireless device for the first transmission direction and a sixth

indication of the second section of the surface of the wireless device for the second transmission direction. The wireless device **506** may include means for selecting the first section of the surface of the wireless device for the first transmission direction and the second section of the surface of the wireless device for the second transmission direction based on the fifth indication and the sixth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration. The first transmissive coefficient value may be associated with the first transmission direction and the second transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction and the second transmission direction. The wireless device **506** may include means for selecting the entire surface area of the surface of the wireless device as the first section of the surface of the wireless device for the first transmission direction and as the second section of the surface of the wireless device for the second transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction. The wireless device **506** may include means for selecting the entire surface area of the surface of the wireless device as the first section of the surface of the wireless device for the first transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a fifth indication of the first transmission direction. The wireless device **506** may include means for selecting the first section of the surface of the wireless device for the first transmission direction based on the fifth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device. The wireless device **506** may include means for configuring a first transmissive coefficient of a first set of meta-elements of the first section of the surface of the wireless device based on the fourth indication of at least one of the incident angle or the incident distance. The wireless device **506** may include means for configuring a second transmissive coefficient of a second set of meta-elements of the second section of the surface of the wireless device based on the fifth indication of at least one of the transmissive angle or the transmissive distance. The incident angle may be equal to the transmissive angle. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle and a seventh indication of at least one of a minimum incident angle or a maximum incident angle. The wireless device **506** may include means for configuring the first transmissive coefficient of the first set of meta-elements of the first section of the surface of the wireless device to perform beam sweeping based on at least one of the minimum transmissive angle or the maximum transmissive angle. The wireless device **506** may include means for configuring the second transmissive coefficient of the second set of meta-elements of the second section of the surface of the wireless device to perform the beam sweeping based on at least one of the minimum incident angle or the maximum incident angle. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a first beamforming gain factor. The wireless device **506** may include means for applying the first beamforming gain factor to at least one meta-element of the surface of the wireless device. The wireless device **506** may include means for receiving, from the network node **502**, a second

transmissive signal configuration or a second transmissive and reflective configuration including a second beamforming gain factor. The wireless device **506** may include means for applying the second beamforming gain factor to the at least one meta-element of the surface of the wireless device. The wireless device **506** may include means for applying the second beamforming gain factor and the first beamforming gain factor to the at least one meta-element of the surface of the wireless device. The third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with a set of meta-elements of the surface of the wireless device is fixed. The third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with meta-elements of the surface of the wireless device is configurable. The transmissive and reflective configuration may include a fourth indication of a power split ratio between the transmission and the reflection. The wireless device **506** may include means for configuring a set of meta-elements of the surface of the wireless device based on the fourth indication of the power split ratio between the transmission and the reflection. The third section and the fourth section may include an overlapping section of the surface of the wireless device. The wireless device **506** may include means for selecting the third section of the surface of the wireless device for the transmissive mode by selecting a first set of time periods for the overlapping section of the surface of the wireless device. The wireless device **506** may include means for selecting the fourth section of the surface of the wireless device for the reflective mode by selecting a second set of time periods for the overlapping section of the surface of the wireless device. The first set of time periods and the second set of time periods may be non-overlapping. The wireless device **506** may include means for configuring at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based on the transmissive and reflective configuration. The wireless device **506** may include a RIS. The first section and the second section may overlap. The means may be the penetrative signal component **198** of the wireless device **506** configured to perform the functions recited by the means.

[0089] FIG. **6A** is a diagram **600** illustrating an example of meta-elements **606** of a wireless device **608** configured to transmit a set of signals **611** from a network node **602** to a UE **604** as the transmissive signals **612**. The network node **602** and the UE **604** may both be in the far field of the surface of the wireless device **608** including the multiple meta-elements **606**. When a signal is transmitted from the network node **602** to the wireless device **608** at the incident angle $\theta_{\text{sub},i}$, the equivalent channel response value of the n th element of the wireless device **608** at a transmissive angle $\theta_{\text{sub},t}$ may be represented by

$$[00001] h_n = e^{j^2 \frac{d_n}{\lambda} (\sin \theta_{\text{sub},i} + \sin \theta_{\text{sub},t})} \times \alpha_{\text{sub},n} e^{j \phi_{\text{sub},n}}.$$

[0090] $\alpha_{\text{sub},n}$ may be the transmissive coefficient of the meta-element n , for example the meta-element **606** may be the first meta-element of a total of N elements.

[0091] $d_{\text{sub},n}$ may be the distance between the n th meta-element to the first meta-element.

[0092] j may be a complex value symbol.

[0093] λ may be the wavelength of the signal reflected off of the meta-element n .

[0094] $\phi_{\text{sub},n}$ may be an amplitude of a reflection coefficient at the n th meta-element. $\phi_{\text{sub},n}$ may be a phase of the transmissive coefficient at the n th meta-element.

[0095] The overall equivalent channel response value of all of the elements of the wireless device **608** at the transmissive angle $\theta_{\text{sub},t}$ may be represented by:

$$[00002] h = \sum_{n=1}^N h_n = \sum_{n=1}^N e^{j^2 \frac{d_n}{\lambda} (\sin \theta_{\text{sub},i} + \sin \theta_{\text{sub},t})} \times \alpha_{\text{sub},n} e^{j \phi_{\text{sub},n}}$$

[0096] If the transmissive coefficient satisfies $\alpha_{\text{sub},n} = \alpha$, then the value of h_n may be estimated as

$$[00003] h_n = e^{-j^2 \frac{d_n}{\lambda} (\sin \theta_{\text{sub},i} + \sin \theta_{\text{sub},t})}$$

[0097] The transmissive beam may point to the direction $\theta_{\text{sub},t}$.

[0098] The coefficient amplitude and phase values of each of the meta-elements **606** of the wireless device **608** may be obtained from a limited set $\{(\alpha_{\text{sub},1}, \phi_{\text{sub},1}), (\alpha_{\text{sub},2}, \phi_{\text{sub},2}), \dots\}$.

$(\alpha_{\text{sub},m}, \phi_{\text{sub},m})$ by different configurations, where $\alpha_{\text{sub},m}$ may be the amplitude of the m th candidate transmissive coefficient and $\phi_{\text{sub},m}$ may be the phase of the m th candidate transmissive coefficient. In other words, the actual beam shape may deviate from the ideal estimated beam direction $\theta_{\text{sub},t}$. The larger the number of meta-elements **606** of the wireless device **608**, the closer the actual beam shape may be to the ideal beam, which may increase the accuracy of the estimated beam direction θ_t .

[0099] FIG. **6B** is a diagram **650** illustrating another example of meta-elements **656** of a wireless device **658** configured to transmit a set of signals **661** from a network node **602** to a UE **604** as the transmissive signals **662**. The network node **602** and the UE **604** may both be in the near field of the surface of the wireless device **658** including the multiple meta-elements **656**. As a result, the incident angle $\theta_{\text{sub},i,n}$ and the transmissive angle $\theta_{\text{sub},t,n}$ at each meta-element n may be different than the incident angles and transmissive angles associated with at least one other meta-element of the wireless device **658**. The incident angle $\theta_{\text{sub},i,n}$ and the transmissive angle $\theta_{\text{sub},t,n}$ at each meta-element n may be calculated based on the meta-element position, the orientation of the surface of the wireless device **658**, and the position of the Tx beam and the Rx beam relative to the surface of the wireless device **658**.

[0100] The selected coefficient amplitude $\alpha_{\text{sub},n}$ and phase $\phi_{\text{sub},n}$ of each meta-element n may be related to the incident angle $\theta_{\text{sub},i,n}$ and the transmissive angle $\theta_{\text{sub},t,n}$, which may depend on both the direction and the distance of the network node **602** and the UE **604** relative to the wireless device **658**.

[0101] When a signal is transmitted towards a meta-element n at an incident angle $\theta_{\text{sub},i,n}$, the equivalent channel response value of the n th element of the wireless device **658** may, again, be represented by

$$[00004] h_n = e^{j^2 \frac{d_n}{2} (\sin \theta_{i,n} + \sin \theta_{t,n})} \times \alpha_n e^{j \phi_n}.$$

[0102] Similarly, the overall equivalent channel response value of all of the elements of the wireless device **658** at all of the transmissive angles $\theta_{\text{sub},t}$ may be represented by:

$$[00005] h = \sum_{n=1}^N h_n = \sum_{n=1}^N e^{j^2 \frac{d_n}{2} (\sin \theta_{i,n} + \sin \theta_{t,n})} \times \alpha_n e^{j \phi_n}$$

[0103] Similarly, if the transmissive coefficient satisfies $\alpha_{\text{sub},n} \equiv \alpha$, then the value of $\phi_{\text{sub},n}$ may be estimated as

$$[00006] \phi_n = -\frac{2}{\alpha} \frac{d_n}{2} (\sin \theta_{i,n} + \sin \theta_{t,n})$$

[0104] The transmissive beam may point generally to the direction of all of the transmissive angles $\{\theta_{\text{sub},t,n}\}$.

[0105] The wireless device **658** may determine the selected coefficient amplitude α_n and phase $\phi_{\text{sub},n}$ of each meta-element n by selecting one available configuration whose amplitude and phase are closest to the theoretical values.

[0106] FIG. **7A** is a diagram **700** illustrating an example of a wireless device **704** configured to assist a network node **702** in performing monostatic sensing of a target object **708**. The wireless device **704** may provide a window between a blocking object **706** that does not allow wireless signals from the network node **702** to penetrate. In other words, the wireless device **704** may be a transmissive object that may be used for a radar (e.g., a network node or a UE) outside an enclosed space to sense an object within the enclosed space. Such a wireless device may be used for a plurality of purposes, such as indoor security (e.g., intruder detection), safety (e.g., monitoring a child within an enclosed space), behavior detection (e.g., determining if a person is walking, running, sitting, or standing), or status detection (e.g., determining if a room is full or empty). If the wireless device **704** is set to a reflective mode, the network node **702** may be relatively far from the boresight direction. The incident angle $\theta_{\text{sub},i}$, $\theta_{\text{sub},i,n}$, may be much larger than zero in such aspects. If the wireless device **704** is set to a transmissive mode, the network node **702** may be within or near the boresight direction. The incident angle $\theta_{\text{sub},i}$, $\theta_{\text{sub},i,n}$, may be equal to or slightly larger than zero in such aspects. Since the coefficient gain of each meta-element may be

proportional to $\cos \theta$, the transmissive signal transmitted by a wireless device set to a transmissive mode may have more meta-element radiation power than a reflective signal reflected by a wireless device set to a reflective mode.

[0107] In some aspects, the wireless device **704** may be configured to have transmissive coefficient values for two transmission (penetration) directions that are identical. In other aspects, the wireless device **704** may be configured to have transmissive coefficient values that are different. The network node **702** may be configured to receive such information from the wireless device **704** (e.g., whether the penetration direction is relevant or irrelevant to the transmissive coefficient value) in order to configure the wireless device **704** correctly.

[0108] In some aspects, the transmissive beamforming gain of the wireless device **704** may depend upon the number of meta-elements at the surface of the wireless device **704** and the beam width of the sensing beam transmitted by the wireless device transmitting the sensing signal. The wireless device **704** may be configured to adjust its transmissive beamforming gain by controlling the number of active meta-elements and the beam width of the sensing beam. For example, the higher the beamforming gain, the larger the received signal strength may be. In another example, the larger the beam width, the smaller amount of radio resources may be. Thus, the network node **702** may be configured to control whether and how much to adjust the transmissive beamforming gain.

[0109] In some aspects, the wireless device **704** may be configured to simultaneously redirect a signal and reflect a signal, for example by redirecting a portion of a signal and reflecting a portion of the signal. The network node **702** may be configured to manage the surface split, incident angles, transmissive angles, reflective angles, and proper power split ratio between transmission and reflection.

[0110] The wireless device **704** may be configured to transmit a capability report message to the network node **702** indicating its capability of bidirectional transmission. For example, the capability report may indicate whether an element of the wireless device may be configured to reflect a beam in a direction, redirect a beam in a direction, or reflect a portion of a beam in one direction and redirect another portion of the beam in another direction. The capability report may also indicate whether the wireless device **704** may configure one element to reflect/redirect a beam in one direction, and another element to reflect/redirect a beam in an opposing direction (e.g., whether the surface may be configured for monodirectional reflection/redirect or whether the surface may be configured for bidirectional reflection/redirect.) The wireless device **704** may indicate for each meta-element configuration (e.g., electrical voltage), whether the transmissive coefficient values for two transmission directions are identical or different. In some aspects, the wireless device **704** may also report whether it may support simultaneous bidirectional transmission. Such a report may be used in aspects where simultaneous bidirectional transmission is not a standard capability.

[0111] In response to receiving the report, the network node **702** may transmit a message to the wireless device **704** to configure transmission directions and surface split based on a sensing type. If the sensing signal transmitter and the receiver are on the same side of the wireless device **704**, the network node **702** may act as a monostatic radar. If, for a certain configuration (e.g., an electrical voltage), the meta-element transmissive coefficient value is irrelevant to the incident angle, the network node **702** may configure the wireless device **704** to use the entire meta-surface of the wireless device **704** (i.e., all of its meta-elements) for the two transmission directions. In such an aspect, the network node **702** may transmit a sensing signal **710** to the wireless device **704**, the wireless device may redirect the sensing signal **710** as the transmissive signal **711** at the target object **708**, the target object **708** may then reflect the transmissive signal **711** as the reflected signal **712** back towards the wireless device **704**, and the wireless device **704** may then redirect the reflected signal **712** towards the network node **702** as the redirected reflected signal **713**. The network node **702** may then perform sensing on the redirected reflected signal **713** to determine attributes of the target object **708**. In some aspects, the network node **702** may transmit a sensing

report to a sensing entity, such as an LMF.

[0112] FIG. 7B is a diagram 730 illustrating an example of the wireless device 704 configured to assist the network node 702 in performing bistatic sensing of the target object 708 with the UE 703. The UE 703 may be located opposite the network node 702 relative to the blocking object 706.

[0113] In response to the sensing signal transmitter (e.g., the network node 702) and the sensing signal receiver (e.g., the UE 703) being on two opposing sides of the wireless device 704, the network node 702 may be configured to act as the transmitter of a bistatic radar while the UE 703 (or another network node) may be configured to act as the receiver of the bistatic radar. The network node 702 may configure the wireless device 704 to use the entire meta-surface for a transmission direction, for example from the side with the network node 702 to the side with the UE 703 or vice-versa. In some aspects, the configuration may include the transmission direction. For example, in one aspect the network node 702 may transmit a configuration to the wireless device 704 to use the entire meta-surface for transmitting a signal from the side with the network node 702 to the side with the UE 703, or to use the entire meta-surface for transmitting a signal from the side with the UE 703 to the side with the network node 702, or to use a first section of the meta-surface for transmitting a first signal from the side with the network node 702 to the side with the UE 703 and a second section of the meta-surface for transmitting a second signal from the side with the UE 703 to the side with the network node 702. The network node 702 may indicate the first section and the second section in a plurality of ways, for example by designating percentages to each section (e.g., 50% for the first section and 50% for the second section, or 30% for the first section and 70% for the second section) or by designating coordinates to each section (e.g., (0,0), (20,20) for two opposing corners of the first section and (20,0), (40,20) for two opposing corners of the second section).

[0114] The network node 702 may configure the wireless device 704 to use the entire meta-surface of the wireless device 704 (i.e., all of its meta-elements) for a single transmission direction, irrespective of a capability report that may be transmitted to the network node 702 by the wireless device 704. In some aspects, the network node 702 may configure the wireless device 704 to redirect signals in a direction, for example from the side with the network node 702 towards the side with the target object 708. In such an aspect, the network node 702 may transmit a sensing signal 710 to the wireless device 704, the wireless device may redirect the sensing signal 710 as the transmissive signal 711 at the target object 708, and the target object 708 may then reflect the transmissive signal 711 as the reflected signal 744 towards the UE 703. In some aspects, the network node 702 may configure the wireless device 704 to designate at least a portion, or the entire, surface for a transmission direction (e.g., from the side with the network node 702 to the side with the UE 703 or from the side with the UE 703 to the side with the network node 702.). In response to receiving the transmission direction, the wireless device 704 may alter the transmissive coefficient value of one or more elements to realize the configured transmission direction. The UE 703 may perform sensing on the reflected signal 744 to determine attributes of the target object 708. In some aspects, the UE 703 may transmit a sensing report to the network node 702 or another sensing entity, such as an LMF.

[0115] FIG. 7C is a diagram 760 illustrating an example of the wireless device 764 configured to assist the network node 702 in performing monostatic or bistatic sensing of the target object 708 using a plurality of differently configured surfaces, a first section 765 and a second section 767. The network node 702 may perform monostatic sensing by measuring the redirected reflected signal 773 from the wireless device 764, or may perform bistatic sensing by configuring the UE 763 to measure the redirected reflected signal 774 from the wireless device 764.

[0116] The wireless device 764 may be configured to transmit a capability report message to the network node 702 indicating its capability of bidirectional transmission. The wireless device 764 may indicate for each meta-element configuration (e.g., electrical voltage), whether the transmissive coefficient values for two transmission directions are identical or different. In some

aspects, the wireless device **764** may also report whether it may support simultaneous bidirectional transmission. Such a report may be used in aspects where simultaneous bidirectional transmission is not a standard capability.

[0117] In response to receiving the report, the network node **702** may transmit a message to the wireless device **764** to configure transmission directions and surface split based on a sensing type. If the sensing signal transmitter and the receiver are on the same side of the wireless device **764**, the network node **702** may act as a monostatic radar, or the network node **702** and the UE **763** may act as a bistatic radar. If, for a certain configuration (e.g., an electrical voltage), the meta-element transmissive coefficient value is relevant to the incident angle, the network node **702** may configure the wireless device **764** to split the meta-surface into a first section **765** for a first transmission direction and a second section **767** for a second transmission direction. In such an aspect, the network node **702** may transmit a sensing signal **770** to the first section **765** of the surface of the wireless device **764**, the wireless device **764** may redirect the sensing signal **770** as the transmissive signal **771** at the target object **708**, the target object **708** may then reflect the transmissive signal **771** as the reflected signal **772** back towards the second section **767** of the surface of the wireless device **764**, and the wireless device **764** may then redirect the reflected signal **772** towards the network node **702** as the redirected reflected signal **773** and/or towards the UE **763** as the redirected reflected signal **774**. In some aspects, the wireless device **764** may be configured to redirect all of the reflected signal **772** towards the network node **702** as the redirected reflected signal **773**, in other aspects, the wireless device may be configured to redirect all of the reflected signal **772** towards the UE **763** as the redirected reflected signal **774**, and in still other aspects the wireless device may be configured to redirect a first portion of the reflected signal **772** towards the network node **702** as the redirected reflected signal **773** and a second portion of the reflected signal **772** towards the UE **763** as the redirected reflected signal **774**.

[0118] The network node **702** may perform sensing on the redirected reflected signal **773** to determine attributes of the target object **708**. In some aspects, the network node **702** may transmit a sensing report to a sensing entity, such as an LMF. The UE **763** may perform sensing on the redirected reflected signal **774** to determine attributes of the target object **708**. In some aspects, the UE **763** may transmit a sensing report to the network node **702** or to a sensing entity, such as an LMF.

[0119] FIG. **8A** is a diagram **800** illustrating another example of a wireless device **804** configured to assist a network node **802** in performing monostatic sensing of a target object **808** using a plurality of differently configured surfaces of the wireless device **804**—namely a first section **805** of the wireless device **804** and a second section **807** of the wireless device **804**. The wireless device **804** may provide a window between a blocking object **806** that does not allow wireless signals from the network node **802** to penetrate. In other words, the wireless device **804** may be a transmissive object that may be used for a radar (e.g., a network node or a UE) outside an enclosed space to sense an object within the enclosed space.

[0120] The network node **802** may configure the wireless device **804** via a configuration transmitted to the wireless device **804**. The configuration may include an indication of an incident angle $\theta_{\text{sub},i,1}$ of the sensing signal **810** at the surface of the first section **805** or a distance between the network node **802** and the wireless device **804**. The configuration may include an indication of a transmissive angle $\theta_{\text{sub},t,2}$ of the redirected reflected signal **813** at the surface of the second section **807** or a distance between the network node **802** and the wireless device **804**. The wireless device **804** may use the incident angle $\theta_{\text{sub},i,1}$ or the distance between the network node **802** and the wireless device **804** to configure the first section **805** of the wireless device **804** to redirect the sensing signal **810** to a direction, such as the transmissive angle $\theta_{\text{sub},t,1}$ of the transmissive sensing signal **811**, which may be used to sense the target object **808**, or an area about the target object **808**. The wireless device **804** may use the transmissive angle $\theta_{\text{sub},t,2}$ or the distance between the network node **802** and the wireless device **804** to redirect the reflected signal **812** as

the redirected reflected signal **813** back to the network node **802**. Since the network node **802** performs monostatic sensing, the incident angle $\theta_{\text{sub},i,1}$ may equal the transmissive angle $\theta_{\text{sub},t,2}$. In some aspects, the network node **802** may configure the wireless device **804** using the distance between the network node **802** and the wireless device **804** when the network node **802** is in a near field of the wireless device **804**.

[0121] In some aspects, the configuration may include an indication of the wireless device **804** to perform beam sweeping using the sensing signal **810** to sweep the transmissive sensing signal **811** across an area. In other words, the wireless device **804** may change a transmissive coefficient of elements of the surface of the first section **805** of the wireless device **804** to change the transmissive angle $\theta_{\text{sub},t,1}$ of the transmissive sensing signal **811** to cover an area. The wireless device **804** may also change a transmissive coefficient of elements of the surface of the second section **807** of the wireless device **804** to account for the change in the incident angle $\theta_{\text{sub},t,2}$ of the reflected signal **812** to such that the transmissive angle $\theta_{\text{sub},t,2}$ remains constantly focused back at the network node **802** as the wireless device **804** performs the beam sweeping.

[0122] The network node **802** may transmit a sensing signal **810** to the first section **805** of the wireless device **804** at an incident angle $\theta_{\text{sub},i,1}$. The wireless device **804** may redirect the sensing signal **810** as the transmissive sensing signal **811** to the target object **808** at the transmissive angle $\theta_{\text{sub},t,1}$. The target object **808** may reflect the transmissive sensing signal **811** to the second section **807** of the wireless device **804** at the incident angle $\theta_{\text{sub},i,2}$ as the reflected signal **812**. The wireless device **804** may redirect the reflected signal **812** back towards the network node **802** at the transmissive angle $\theta_{\text{sub},t,2}$ as the redirected reflected signal **813**. The network node **802** may then perform sensing on the redirected reflected signal **813** to determine attributes of the target object **808**. In some aspects, the network node **802** may transmit a sensing report to a sensing entity, such as an LMF.

[0123] FIG. **8B** is a diagram **850** illustrating another example of a wireless device **804** configured to assist a network node **802** in performing bistatic sensing of a target object **808** using a plurality of differently configured surfaces—namely a first section **805** of the wireless device **804** and a second section **807** of the wireless device **804**, with a UE **863** acting as a sensing receiver.

[0124] The network node **802** may configure the wireless device **804** via a configuration transmitted to the wireless device **804**. The configuration may include an indication of an incident angle $\theta_{\text{sub},i,1}$ of the sensing signal **810** at the surface of the first section **805** or a distance between the network node **802** and the wireless device **804**. The configuration may include an indication of a transmissive angle $\theta_{\text{sub},t,2}$ of the redirected reflected signal **813** at the surface of the second section **807** or a distance between the UE **863** and the wireless device **804**. The wireless device **804** may use the incident angle $\theta_{\text{sub},t,1}$ or the distance between the network node **802** and the wireless device **804** to configure the first section **805** of the wireless device **804** to redirect the sensing signal **810** to a direction, such as the transmissive angle $\theta_{\text{sub},t,1}$ of the transmissive sensing signal **811**, which may be used to sense the target object **808**, or an area about the target object **808**. The wireless device **804** may use the transmissive angle $\theta_{\text{sub},t,2}$ or the distance between the UE **863** and the wireless device **804** to redirect the reflected signal **812** as the redirected reflected signal **814** to the UE **863**. Since the network node **802** performs bistatic sensing, the incident angle $\theta_{\text{sub},t,1}$ may not be equal the transmissive angle $\theta_{\text{sub},t,2}$. In some aspects, the network node **802** may configure the wireless device **804** using the distance between the network node **802** and the wireless device **804** and the distance between the UE **863** and the wireless device **804** when the network node **802** is in a near field of the wireless device **804** and the UE **863** is in a near field of the wireless device **804**.

[0125] In some aspects, the configuration may include an indication of the wireless device **804** to perform beam sweeping using the sensing signal **810** to sweep the transmissive sensing signal **811** across an area. In other words, the wireless device **804** may change a transmissive coefficient of elements of the surface of the first section **805** of the wireless device **804** to change the

transmissive angle $\theta_{\text{sub.t},2}$ of the transmissive sensing signal **811** to cover an area. The wireless device **804** may also change a transmissive coefficient of elements of the surface of the second section **807** of the wireless device **804** to account for the change in the incident angle $\theta_{\text{sub.i},2}$ of the reflected signal **812** to such that the transmissive angle $\theta_{\text{sub.i},2}$ remains constantly focused back at the UE **863** as the wireless device **804** performs the beam sweeping.

[0126] The network node **802** may transmit a sensing signal **810** to the first section **805** of the wireless device **804** at an incident angle $\theta_{\text{sub.i},1}$. The wireless device **804** may redirect the sensing signal **810** as the transmissive sensing signal **811** to the target object **808** at the transmissive angle $\theta_{\text{sub.t},1}$. The target object **808** may reflect the transmissive sensing signal **811** to the second section **807** of the wireless device **804** at the incident angle $\theta_{\text{sub.i},2}$ as the reflected signal **812**. The wireless device **804** may redirect the reflected signal **812** back towards the UE **863** at the transmissive angle $\theta_{\text{sub.t},2}$ as the redirected reflected signal **814**. The UE **863** may then perform sensing on the redirected reflected signal **814** to determine attributes of the target object **808**. In some aspects, the UE **863** may transmit a sensing report to the network node **802** or a sensing entity, such as an LMF.

[0127] FIG. 9A is a diagram **900** illustrating another example of a wireless device **904** configured to assist a network node **902** in performing monostatic sensing of a target object **908**. The wireless device **904** may be configured to generate a transmissive beamforming gain equal to

[00007] $\frac{g_{\text{max}}}{\beta}$,

where $g_{\text{sub.max}}$ may be the maximum transmissive beamforming gain with the narrowest beam, and β may be a beamforming gain factor. In some aspects, the possible values of β may be [1, 2, 3, 4, ...] or [1, 2, 4, 8, ...]. In other words, the values of β may be consecutive integers, or may be doubled for each subsequent value of β .

[0128] The wireless device **904** may provide a window between a blocking object **906** that does not allow wireless signals from the network node **902** to penetrate. In other words, the wireless device **904** may be a transmissive object that may be used for a radar (e.g., a network node or a UE) outside an enclosed space to sense an object within the enclosed space.

[0129] The network node **902** may transmit a sensing signal **910** to the first section **905** of the wireless device **904**. The wireless device **904** may redirect the sensing signal **910** as the transmissive sensing signal **911** to the target object **908**. The wireless device **904** may redirect the sensing signal **910** by applying an initial beamforming factor of $\beta=1$ to form a narrow beam **916** towards the target object **908**. The target object **908** may reflect the transmissive sensing signal **911** to the second section **907** of the wireless device **904** as the reflected signal **912**. The wireless device **904** may redirect the reflected signal **912** back towards the network node **902** as the redirected reflected signal **913**. The network node **902** may then perform sensing on the redirected reflected signal **913** to determine attributes of the target object **908**. In some aspects, the network node **902** may transmit a sensing report to a sensing entity, such as an LMF.

[0130] Based on the received sensing signal strength of the redirected reflected signal **913**, the network node **902** may dynamically reconfigure the beamforming gain factor β so that the new transmissive beamforming gain is increased or decreased.

[0131] FIG. 9B is a diagram **950** illustrating an example of the wireless device **904** configured to use a new beamforming factor of $\beta=2$ to form a wide beam **918**. The network node **902** may transmit a sensing signal **910** to the first section **905** of the wireless device **904**. The wireless device **904** may redirect the sensing signal **910** as the transmissive sensing signal **961** to the target object **908**. The wireless device **904** may redirect the sensing signal **910** by applying an updated beamforming factor of $\beta=2$ to form a wide beam **918** towards the target object **908**. The target object **908** may reflect the transmissive sensing signal **961** to the second section **907** of the wireless device **904** as the reflected signal **962**. The wireless device **904** may redirect the reflected signal **962** back towards the network node **902** as the redirected reflected signal **963**. The network node

902 may then perform sensing on the redirected reflected signal **963** to determine attributes of the target object **908**. In some aspects, the network node **902** may transmit a sensing report to a sensing entity, such as an LMF.

[0132] The network node **902** may transmit a configuration of the sensing signal **910** to the wireless device **904** to apply the beamforming gain factor β to the sensing signal **910**. The network node **902** may dynamically reconfigure the beamforming gain factor β to increase or decrease the beamforming applied by the wireless device **904**. In one aspect, the network node **902** may initially transmit a beamforming gain factor β to the wireless device **804**. Later, the network node **902** may indicate a new beamforming gain factor as an absolute value. In other words, the network node **902** may initially transmit a beamforming gain factor $\beta_{\text{sub.1}}$, such that the transmissive beamforming gain is

$$[00008] g_1 = \frac{g_{\text{max}}}{1},$$

and then may transmit a beamforming gain factor $\beta_{\text{sub.2}}$, such that the transmissive beamforming gain is

$$[00009] g_2 = \frac{g_{\text{max}}}{2}.$$

In another aspect, the network node **902** may initially transmit a beamforming gain factor β to the wireless device **804**. Later, the network node **902** may indicate a new beamforming gain factor as a relative change value from the previous beamforming gain factor. In other words, the network node **902** may initially transmit a beamforming gain factor $\beta_{\text{sub.1}}$, such that the transmissive beamforming gain is

$$[00010] g_1 = \frac{g_{\text{max}}}{1},$$

and then may transmit a relative change value in the beamforming gain factor $\Delta_{\text{sub.p}}$, such that the transmissive beamforming gain is

$$[00011] g_2 = \frac{g_{\text{max}}}{1 + \Delta_{\text{sub.p}}}.$$

[0133] FIG. **10A** is a diagram **1000** illustrating an example of a wireless device **1004** configured to assist a network node **1002** in performing monostatic sensing of a target object **1008** opposite the network node **1002** and a target object **1009** on the same side as the network node **1002**. In other words, the network node **1002** may sense areas located on both sides of the wireless device **1004** (e.g., an indoor area and an outdoor area, or two neighbor rooms separated by a wall). The wireless device **1004** may be configured to use both transmissive and reflective properties of its meta-elements to achieve this. The wireless device **1004** may provide a window between a blocking object **1006** that does not allow wireless signals from the network node **1002** to penetrate. In other words, the wireless device **1004** may be a transmissive object that may be used for a radar (e.g., a network node or a UE) outside an enclosed space to sense an object within the enclosed space.

[0134] The wireless device **1004** may transmit a capability report message to the network node **1002**. The capability report message may indicate whether and how to support simultaneous transmissive and reflective properties of its elements. The capability report message may indicate a capability type of a set of elements of the wireless device **1004**. A first capability type may be associated with a set of elements that are configured to support either transmissive signals or reflective signals. A second capability type may be associated with a set of elements that are configured to support both transmissive signals and reflective signals, where the power may have a fixed split between transmission and reflection (e.g., half of the signal is transmitted, and half of the signal is reflected). A third capability type may be associated with a set of elements that are configured to support both transmissive signals and reflective signals, where the power split between transmission and reflection may be configurable. In some aspects, the wireless device **1004** may have a plurality of selectable splits, which the wireless device **1004** may indicate to the network node **1002** (e.g., a first power split ratio of $\frac{1}{3}$ transmissive: $\frac{2}{3}$ reflective, a second power split ratio of $\frac{1}{2}$ transmissive: $\frac{1}{2}$ reflective, and a third power split ratio of $\frac{2}{3}$ transmissive: $\frac{1}{3}$

reflective). The network node **1002** may configure the wireless device **1004** in response to the capability report message. For example, if the capability report message indicates that a set of elements may be configured to be either transmissive or reflective, the network node **1002** may configure a first portion of the surface to be transmissive and a second portion of the surface to be reflective, for example half of the surface for transmissive signals and half of the surface for reflective signals, or 40% of the surface for transmissive signals and 60% of the surface for reflective signals. In some aspects, the network node **1002** may configure a time occasion pattern, such that in one time occasion the wireless device **1004** is configured for a transmissive signal, and in another time occasion the wireless device **1004** is configured for a reflective signal. The network node **1002** may indicate a time length for the transmissive signal and a time length for the reflective signal, and the wireless device **1004** may alternate each configuration accordingly.

[0135] If the capability report message indicates that a set of elements may be configured to be both transmissive or reflective, the network node **1002** may configure the surface of the wireless device **1004** to be both transmissive and reflective. The network node **1002** may transmit an indication of the incident angle for the sensing signal **1010**, the transmissive angle for the redirected reflected signal **1013**, and the reflective angle for the reflected signal **1017** to the wireless device **1004**, and may indicate for the wireless device **1004** to sweep beams across an area on one side of the wireless device **1004** (for the transmissive signal **1011**) and sweep beams across an area of the other side of the wireless device **1004** (for the reflective signal **1015**) while redirecting the redirected reflected signal **1013** and the reflected signal **1017** back to the network node **1002**. Based on the indicated angles, the wireless device **1004** may configure its meta-elements to generate both the transmissive and reflective coefficients such that both the transmissive and reflective beams aim at the respective target directions. In other words, the transmissive signal **1011** aims at the target object **1008**, or an area about the target object **1008**, and the reflective signal **1015** aims at the target object **1009**, or an area about the target object **1009**. In some aspects, the coefficient value may be irrelevant to the incident angle (e.g., at a certain electrical voltage of the wireless device **1004**). The wireless device **1004** may report this capability to the network node **1002**. In response, the network node **1002** may configure the wireless device **1004** to use its entire surface for both transmissive and reflective sensing signals.

[0136] If the capability report message indicates that a set of elements may be configured to be both transmissive or reflective with a configurable power split, the network node **1002** may indicate the power split ratio to use. In some aspects, the network node **1002** may configure the power split based on a distance to a target object or a size of a target object. For example, if the distance between the wireless device **1004** and the target object **1008** is greater than the distance between the wireless device **1004** and the target object **1009**, the network node **1002** may configure more power to the transmissive signal (e.g., select a power split ratio such that the transmissive signal **1011** is stronger than the reflective signal **1015**). In another example, if the size of the target object **1009** is smaller than the size of the target object **1008**, the network node **1002** may configure more power to the reflective signal (e.g., select a power split ratio such that the reflective signal **1015** is stronger than the transmissive signal **1011**).

[0137] The network node **1002** may transmit a sensing signal **1010** to the wireless device **1004**, the wireless device may redirect a portion of the sensing signal **1010** as the transmissive signal **1011** at the target object **1008**, the target object **1008** may then reflect the transmissive signal **1011** as the reflected signal **1012** back towards the wireless device **1004**, and the wireless device **1004** may then redirect the reflected signal **1012** towards the network node **1002** as the redirected reflected signal **1013**. The network node **1002** may then perform sensing on the redirected reflected signal **1013** to determine attributes of the target object **1008**. In some aspects, the network node **1002** may transmit a sensing report to a sensing entity, such as an LMF.

[0138] The wireless device may also reflect a portion of the sensing signal **1010** as the reflective signal **1015** at the target object **1009**, the target object **1009** may then reflect the reflective signal

1015 as the reflected signal **1016** back towards the wireless device **1004**, and the wireless device **1004** may then reflect the reflected signal **1016** towards the network node **1002** as the reflected signal **1017**. The network node **1002** may then perform sensing on the reflected signal **1017** to determine attributes of the target object **1009**. In some aspects, the network node **1002** may transmit a sensing report to a sensing entity, such as an LMF.

[0139] FIG. **10B** is a diagram **1050** illustrating an example of a wireless device **1054** configured to assist a network node **1002** in performing monostatic sensing of a target object **1008** opposite the network node **1002** and a target object **1009** on the same side as the network node **1002** using a first section **1055** of the surface of the wireless device **1054** and a second section **1057** of the surface of the wireless device **1054**. The wireless device **1054** may be configured to use transmissive properties of the first section **1055** of the wireless device **1054** and reflective properties of the second section **1057** of the wireless device **1054** to achieve this.

[0140] The network node **1002** may transmit a sensing signal **1060** to the first section **1055** of the wireless device **1004**, the first section **1055** of the wireless device **1054** may redirect a portion of the sensing signal **1060** as the transmissive signal **1061** at the target object **1008**, the target object **1008** may then reflect the transmissive signal **1061** as the reflected signal **1062** back towards the first section **1055** of the wireless device **1054**, and the first section **1055** of the wireless device **1054** may then redirect the reflected signal **1062** towards the network node **1002** as the redirected reflected signal **1063**. The network node **1002** may then perform sensing on the redirected reflected signal **1063** to determine attributes of the target object **1008**. In some aspects, the network node **1002** may transmit a sensing report to a sensing entity, such as an LMF.

[0141] The second section **1057** of the wireless device may also reflect a portion of the sensing signal **1060** as the reflective signal **1065** at the target object **1009**, the target object **1009** may then reflect the reflective signal **1065** as the reflected signal **1066** back towards the second section **1057** of the wireless device **1004**, and the second section **1057** of the wireless device **1004** may then reflect the reflected signal **1066** towards the network node **1002** as the reflected signal **1067**. The network node **1002** may then perform sensing on the reflected signal **1067** to determine attributes of the target object **1009**. In some aspects, the network node **1002** may transmit a sensing report to a sensing entity, such as an LMF.

[0142] FIG. **11** is a connection flow diagram **1100** illustrating an example of communications between a network node **1102** and a wireless device **1104** configured to assist the network node **1102** in performing sensing on a target object **1106**. The wireless device **1104** may transmit a capability report **1108** to the network node **1102**. The capability report **1108** may indicate whether and how to support simultaneous transmission (penetration) and reflection. At **1110**, the network node **1102** may configure the wireless device **1104**. The network node **1102** may transmit a transmissive signal configuration or a transmissive and reflective configuration **1114** to the wireless device **1104**. At **1116**, the wireless device **1104** may apply the received configuration to meta-elements at its surface. The network node **1102** may transmit a sensing signal **1118** to the wireless device **1104**. The wireless device **1104** may transmit and/or reflect portions of the sensing signal **1118** as the transmissive/reflective sensing signal **1120** at the target object **1106**. The target object **1106** may reflect the received signal as the sensing signal **1122** at the wireless device **1104**. The wireless device **1104** may transmit and/or reflect portions of the sensing signal **1122** at the network node **1102** as the transmissive/reflective sensing signal **1124**. At **1126**, the network node **1102** may reconfigure the wireless device **1104** based on measurements of the transmissive/reflective sensing signal **1124**.

[0143] FIG. **12** is a flowchart **1200** of a method of wireless communication. The method may be performed by a wireless device (e.g., the wireless device **106**, the wireless device **506**, the wireless device **608**, the wireless device **658**, the wireless device **704**, the wireless device **804**, the wireless device **904**, the wireless device **1004**, the wireless device **1104**). At **1202**, the wireless device may transmit, for a network node, a capability report including at least one of: (1) a first indication of a

first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. For example, **1202** may be performed by the wireless device **1104** in FIG. **11**, which may transmit, for the network node **1002**, a capability report **1108** including at least one of: (1) a first indication of a first transmissive coefficient value for the surface of the wireless device **1004**, (2) an indication of its sensing capability, or (3) an indication of a transmissive and reflective capability. Moreover, **1202** may be performed by the component **198** in FIG. **1** or **5**.

[0144] At **1204**, the wireless device may receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. For example, **1204** may be performed by the wireless device **1104** in FIG. **11**, which may receive, from the network node **1102**, at least one of a transmissive signal configuration or a transmissive and reflective configuration **1114** based on the capability report **1108**. Moreover, **1204** may be performed by the component **198** in FIG. **1** or **5**.

[0145] At **1206**, the wireless device may select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. For example, **1206** may be performed by the wireless device **1104** in FIG. **11**, which may select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration **1114**, at least one of: (1) a first section of a surface of the wireless device **1004** for a first transmission direction, (2) a second section of the surface of the wireless device **1004** for a second transmission direction, (3) a third section of the surface of the wireless device **1004** for a transmissive mode, or (4) a fourth section of the surface of the wireless device **1004** for a reflective mode. Moreover, **1206** may be performed by the component **198** in FIG. **1** or **5**.

[0146] FIG. **13** is a flowchart **1300** of a method of wireless communication. The method may be performed by a network node (e.g., the base station **102**, the base station **310**; the wireless node **402**, the wireless node **404**, the wireless node **406**, the wireless node **408**; the network node **502**, the network node **602**, the network node **702**, the network node **802**, the network node **902**, the network node **1002**, the network node **1102**; the network entity **1402**, the network entity **1560**). At **1302**, the network node may receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. For example, **1302** may be performed by the network node **1102** in FIG. **11**, which may receive, from the wireless device **1104**, a capability report **1108** including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. Moreover, **1302** may be performed by the component **199** in FIG. **1**, **3**, **14**, or **15**.

[0147] At **1304**, the network node may transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. For example, **1304** may be performed by the network node **1102** in FIG. **11**, which may transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section

of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. Moreover, **1304** may be performed by the component **199** in FIG. **1**, **3**, **14**, or **15**.

[0148] FIG. **14** is a diagram **1400** illustrating an example of a hardware implementation for a network entity **1402**. The network entity **1402** may be a BS, a component of a BS, or may implement BS functionality. The network entity **1402** may include at least one of a CU **1410**, a DU **1430**, or an RU **1440**. For example, depending on the layer functionality handled by the component **199**, the network entity **1402** may include the CU **1410**; both the CU **1410** and the DU **1430**; each of the CU **1410**, the DU **1430**, and the RU **1440**; the DU **1430**; both the DU **1430** and the RU **1440**; or the RU **1440**.

[0149] The CU **1410** may include a CU processor **1412**. The CU processor **1412** may include on-chip memory **1412'**. In some aspects, the CU **1410** may further include additional memory modules **1414** and a communications interface **1418**. The CU **1410** communicates with the DU **1430** through a midhaul link, such as an F1 interface. The DU **1430** may include a DU processor **1432**. The DU processor **1432** may include on-chip memory **1432'**. In some aspects, the DU **1430** may further include additional memory modules **1434** and a communications interface **1438**. The DU **1430** communicates with the RU **1440** through a fronthaul link. The RU **1440** may include an RU processor **1442**. The RU processor **1442** may include on-chip memory **1442'**. In some aspects, the RU **1440** may further include additional memory modules **1444**, one or more transceivers **1446**, antennas **1480**, and a communications interface **1448**. The RU **1440** communicates with the UE **104**. The on-chip memory **1412'**, **1432'**, **1442'** and the additional memory modules **1414**, **1434**, **1444** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the processors **1412**, **1432**, **1442** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

[0150] As discussed supra, the component **199** may be configured to receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The component **199** may be configured to transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The component **199** may be within one or more processors of one or more of the CU **1410**, DU **1430**, and the RU **1440**. The component **199** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. The network entity **1402** may include a variety of components configured for various functions. In one configuration, the network entity **1402** may include means for receiving, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The network entity **1402** may include means for transmitting, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second

section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The capability report may include a fourth indication of a bidirectional transmission capability. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is capable of simultaneous bidirectional transmission. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is not capable of the simultaneous bidirectional transmission. The capability report may include a fourth indication of whether the transmissive coefficient value is relevant to an incident angle of a transmitted signal. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive coefficient value is relevant to the incident angle of the transmitted signal. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive coefficient value is not relevant to the incident angle of the transmitted signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle for beam sweeping a sensing signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a seventh indication of at least one of a minimum incident angle or a maximum incident angle for beam sweeping the sensing signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a first beamforming gain factor associated with a set of meta-elements of the surface of the wireless device. The network entity **1402** may include means for transmitting a sensing signal to the wireless device. The network entity **1402** may include means for transmitting a sensing signal to the wireless device. The network entity **1402** may include means for transmitting at least one of a second transmissive signal configuration or a second transmissive and reflective configuration including a second beamforming gain factor based on the received reflected signal. At least one of the second transmissive signal configuration or the second transmissive and reflective configuration may include at least one of an absolute gain value or a relative gain value relative to the first beamforming gain factor. The third indication of the transmissive and reflective capability may indicate that a power split between transmission and

reflection associated with a set of meta-elements of the surface of the wireless device is configurable. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a power split ratio between the transmission and the reflection. The third section and the fourth section may include an overlapping section of the surface of the wireless device. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a first set of time periods for the overlapping section of the surface of the wireless device and a second set of time periods for the overlapping section of the surface of the wireless device. The first set of time periods and the second set of time periods may be non-overlapping. The network entity **1402** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based. The wireless device may include a RIS. The first section and the second section may overlap. The means may be the component **199** of the network entity **1402** configured to perform the functions recited by the means. As described supra, the network entity **1402** may include the Tx processor **316**, the Rx processor **370**, and the controller/processor **375**. As such, in one configuration, the means may be the Tx processor **316**, the Rx processor **370**, and/or the controller/processor **375** configured to perform the functions recited by the means.

[0151] FIG. **15** is a diagram **1500** illustrating an example of a hardware implementation for a network entity **1560**. In one example, the network entity **1560** may be within the core network **120**. The network entity **1560** may include a network processor **1512**. The network processor **1512** may include on-chip memory **1512'**. In some aspects, the network entity **1560** may further include additional memory modules **1514**. The network entity **1560** communicates via the network interface **1580** directly (e.g., backhaul link) or indirectly (e.g., through a RIC) with the CU **1502**. The on-chip memory **1512'** and the additional memory modules **1514** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. The processor **1512** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

[0152] As discussed supra, the component **199** may be configured to receive, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The component **199** may be configured to transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The component **199** may be within the processor **1512**. The component **199** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. The network entity **1560** may include a variety of components configured for various functions. In one configuration, the network entity **1560** may include means for receiving, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The network entity

1560 may include means for transmitting, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode. The capability report may include a fourth indication of a bidirectional transmission capability. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is capable of simultaneous bidirectional transmission. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is not capable of the simultaneous bidirectional transmission. The capability report may include a fourth indication of whether the transmissive coefficient value is relevant to an incident angle of a transmitted signal. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive coefficient value is relevant to the incident angle of the transmitted signal. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive coefficient value is not relevant to the incident angle of the transmitted signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle for beam sweeping a sensing signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a seventh indication of at least one of a minimum incident angle or a maximum incident angle for beam sweeping the sensing signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a first beamforming gain factor associated with a set of meta-elements of the surface of the wireless device. The network entity **1560** may include means for transmitting a sensing signal to the wireless device. The network entity **1560** may include means for transmitting a sensing signal to the wireless device. The network entity **1560** may include means for transmitting at least one of a second transmissive signal configuration or a second transmissive and reflective configuration including a second beamforming gain factor based on the received reflected signal. At least one of the second

transmissive signal configuration or the second transmissive and reflective configuration may include at least one of an absolute gain value or a relative gain value relative to the first beamforming gain factor. The third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with a set of meta-elements of the surface of the wireless device is configurable. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a power split ratio between the transmission and the reflection. The third section and the fourth section may include an overlapping section of the surface of the wireless device. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a first set of time periods for the overlapping section of the surface of the wireless device and a second set of time periods for the overlapping section of the surface of the wireless device. The first set of time periods and the second set of time periods may be non-overlapping. The network entity **1560** may include means for configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based. The wireless device may include a RIS. The first section and the second section may overlap. The means may be the component **199** of the network entity **1560** configured to perform the functions recited by the means.

[0153] It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.

[0154] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean “one and only one” unless specifically so stated, but rather “one or more.” Terms such as “if,” “when,” and “while” do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., “when,” do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses, or indirectly between the first and second apparatuses through a set of apparatuses. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are

known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

[0155] As used herein, the phrase “based on” shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase “based on A” (where “A” may be information, a condition, a factor, or the like) shall be construed as “based at least on A” unless specifically recited differently.

[0156] A device configured to “output” data, such as a transmission, signal, or message, may transmit the data, for example with a transceiver, or may send the data to a device that transmits the data. A device configured to “obtain” data, such as a transmission, signal, or message, may receive, for example with a transceiver, or may obtain the data from a device that receives the data.

[0157] The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

[0158] Aspect 1 is a method of wireless communication at a UE, where the method may include transmitting, for a network node, a capability report including at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The method may include receiving, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report. The method may include selecting, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

[0159] Aspect 2 is the method of aspect 1, where the capability report may include a fourth indication of whether the first transmissive coefficient value is associated with a monodirectional transmission or a bidirectional transmission.

[0160] Aspect 3 is the method of either of aspects 1 or 2, where the capability report may include a fourth indication of a bidirectional transmission capability.

[0161] Aspect 4 is the method of any of aspects 1 to 3, where the first transmissive coefficient value may be associated with the first transmission direction. The capability report may further include a fourth indication of a second transmissive coefficient value associated with the second transmission direction. The first transmissive coefficient value may be different from the second transmissive coefficient value. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of the first section of the surface of the wireless device for the first transmission direction and a sixth indication of the second section of the surface of the wireless device for the second transmission direction. Selecting the first section of the surface of the wireless device for the first transmission direction and the second section of the surface of the wireless device for the second transmission direction may be based on the fifth indication and the sixth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

[0162] Aspect 5 is the method of any of aspects 1 to 4, where the first transmissive coefficient value may be associated with the first transmission direction and the second transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction and the second transmission direction. The method may include selecting the entire surface area of the surface of the wireless device as the first section of

the surface of the wireless device for the first transmission direction and as the second section of the surface of the wireless device for the second transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

[0163] Aspect 6 is the method of any of aspects 1 to 5, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction. The method may include selecting the entire surface area of the surface of the wireless device as the first section of the surface of the wireless device for the first transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

[0164] Aspect 7 is the method of aspect 6, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a fifth indication of the first transmission direction. Selecting the first section of the surface of the wireless device for the first transmission direction may be based on the fifth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

[0165] Aspect 8 is the method of any of aspects 1 to 7, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device. The method may include configuring a first transmissive coefficient of a first set of meta-elements of the first section of the surface of the wireless device based on the fourth indication of at least one of the incident angle or the incident distance. The method may include configuring a second transmissive coefficient of a second set of meta-elements of the second section of the surface of the wireless device based on the fifth indication of at least one of the transmissive angle or the transmissive distance.

[0166] Aspect 9 is the method of aspect 8, where the incident angle may be equal to the transmissive angle.

[0167] Aspect 10 is the method of either of aspects 8 or 9, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle and a seventh indication of at least one of a minimum incident angle or a maximum incident angle. The method may include configuring the first transmissive coefficient of the first set of meta-elements of the first section of the surface of the wireless device to perform beam sweeping based on at least one of the minimum transmissive angle or the maximum transmissive angle. The method may include configuring the second transmissive coefficient of the second set of meta-elements of the second section of the surface of the wireless device to perform the beam sweeping based on at least one of the minimum incident angle or the maximum incident angle.

[0168] Aspect 11 is the method of any of aspects 1 to 10, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a first beamforming gain factor. The method may include applying the first beamforming gain factor to at least one meta-element of the surface of the wireless device.

[0169] Aspect 12 is the method of any of aspects 1 to 11, where the method may include receiving, from the network node, a second transmissive signal configuration or a second transmissive and reflective configuration including a second beamforming gain factor. The method may include applying the second beamforming gain factor to the at least one meta-element of the surface of the wireless device. The method may include applying the second beamforming gain factor and the first beamforming gain factor to the at least one meta-element of the surface of the wireless device.

[0170] Aspect 13 is the method of any of aspects 1 to 12, where the third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with a set of meta-elements of the surface of the wireless device is fixed.

[0171] Aspect 14 is the method of any of aspects 1 to 13, where the third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with meta-elements of the surface of the wireless device is configurable. The transmissive and reflective configuration may include a fourth indication of a power split ratio between the transmission and the reflection. The method may include configuring a set of meta-elements of the surface of the wireless device based on the fourth indication of the power split ratio between the transmission and the reflection.

[0172] Aspect 15 is the method of any of aspects 1 to 14, where the third section and the fourth section may include an overlapping section of the surface of the wireless device. Selecting the third section of the surface of the wireless device for the transmissive mode may include selecting a first set of time periods for the overlapping section of the surface of the wireless device. Selecting the fourth section of the surface of the wireless device for the reflective mode may include selecting a second set of time periods for the overlapping section of the surface of the wireless device. The first set of time periods and the second set of time periods may be non-overlapping.

[0173] Aspect 16 is the method of any of aspects 1 to 15, where the method may include configuring at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based on the transmissive and reflective configuration.

[0174] Aspect 17 is the method of any of aspects 1 to 16, where the wireless device may include a RIS.

[0175] Aspect 18 is a method of wireless communication at a network node, where the method may include receiving, from a wireless device, a capability report including at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability. The method may include transmitting, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

[0176] Aspect 19 is the method of aspect 18, where the capability report may include a fourth indication of a bidirectional transmission capability. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is capable of simultaneous bidirectional transmission. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is not capable of the simultaneous bidirectional transmission.

[0177] Aspect 20 is the method of either of aspects 18 or 19, where the capability report may include a fourth indication of whether the transmissive coefficient value is relevant to an incident angle of a transmitted signal. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive

coefficient value is relevant to the incident angle of the transmitted signal. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication indicating that the transmissive coefficient value is not relevant to the incident angle of the transmitted signal.

[0178] Aspect 21 is the method of any of aspects 18 to 20, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of the first transmission direction.

[0179] Aspect 22 is the method of any of aspects 18 to 21, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction. At least one of the transmissive signal configuration or the transmissive and reflective configuration may include a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device.

[0180] Aspect 23 is the method of aspect 22, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle for beam sweeping a sensing signal. At least one of the transmissive signal configuration or the transmissive and reflective configuration may further include a seventh indication of at least one of a minimum incident angle or a maximum incident angle for beam sweeping the sensing signal.

[0181] Aspect 24 is the method of any of aspects 18 to 23, where at least one of the transmissive signal configuration or the transmissive and reflective configuration may include a first beamforming gain factor associated with a set of meta-elements of the surface of the wireless device.

[0182] Aspect 25 is the method of any aspect 24, where the method may include transmitting a sensing signal to the wireless device. The method may include receiving a reflected signal from the wireless device based on the transmitted sensing signal and the first beamforming gain factor. The method may include transmitting at least one of a second transmissive signal configuration or a second transmissive and reflective configuration including a second beamforming gain factor based on the received reflected signal.

[0183] Aspect 26 is the method of aspect 25, where at least one of the second transmissive signal configuration or the second transmissive and reflective configuration may include at least one of an absolute gain value or a relative gain value relative to the first beamforming gain factor.

[0184] Aspect 27 is the method of any of aspects 18 to 26, where the third indication of the transmissive and reflective capability may indicate that a power split between transmission and reflection associated with a set of meta-elements of the surface of the wireless device is configurable. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a power split ratio between the transmission and the reflection.

[0185] Aspect 28 is the method of any of aspects 18 to 27, where the third section and the fourth section may include an overlapping section of the surface of the wireless device. The method may include configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate a first set of time periods for the overlapping section of the surface of the wireless device and a second set of time periods for the overlapping section of the surface of the wireless device. The first set of time periods and the second set of time periods may be non-overlapping.

[0186] Aspect 29 is the method of any of aspects 18 to 28, where the method may include

configuring at least one of the transmissive signal configuration or the transmissive and reflective configuration to indicate at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based.

[0187] Aspect 30 is the method of any of aspects 18 to 29, where the wireless device may include a RIS.

[0188] Aspect 31 is the method of any of aspects 8, 9, 10, 22, or 23, where the first section and the second section may overlap. For example, the same transmissive coefficient may be used for bi-directional transmission, such that a RIS may not be split into two sections in order to accommodate both redirection of a first portion of a beam and reflection of a second portion of a beam directed at a surface of the RIS.

[0189] Aspect 32 is an apparatus for wireless communication, including: a memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to implement any of aspects 1 to 31.

[0190] Aspect 33 is the apparatus of aspect 31, further including at least one of an antenna or a transceiver coupled to the at least one processor.

[0191] Aspect 34 is an apparatus for wireless communication including means for implementing any of aspects 1 to 31.

[0192] Aspect 35 is a computer-readable medium (e.g., a non-transitory computer-readable medium) storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 1 to 31.

Claims

1. An apparatus for wireless communication at a wireless device, comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: transmit, for a network node, a capability report comprising at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability; receive, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report; and select, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

2. The apparatus of claim 1, wherein the capability report comprises a fourth indication of whether the first transmissive coefficient value is associated with monodirectional transmission or bidirectional transmission.

3. The apparatus of claim 1, wherein the capability report comprises a fourth indication of a bidirectional transmission capability.

4. The apparatus of claim 1, wherein the first transmissive coefficient value is associated with the first transmission direction, wherein the capability report further comprises a fourth indication of a second transmissive coefficient value associated with the second transmission direction, wherein the first transmissive coefficient value is different from the second transmissive coefficient value, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration comprises a fifth indication of the first section of the surface of the wireless device for the first transmission direction and a sixth indication of the second section of the surface of the wireless device for the second transmission direction, wherein selecting the first section of the surface of the wireless device for the first transmission direction and the second section of the surface of the wireless device for the second transmission direction is based on the fifth indication

and the sixth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

5. The apparatus of claim 1, wherein the first transmissive coefficient value is associated with the first transmission direction and the second transmission direction, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration comprises a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction and the second transmission direction, wherein the at least one processor is further configured to: select the entire surface area of the surface of the wireless device as the first section of the surface of the wireless device for the first transmission direction and as the second section of the surface of the wireless device for the second transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

6. The apparatus of claim 1, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration comprises a fourth indication of an entire surface area of the surface of the wireless device for the first transmission direction, wherein the at least one processor is further configured to: select the entire surface area of the surface of the wireless device as the first section of the surface of the wireless device for the first transmission direction based on the fourth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

7. The apparatus of claim 6, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration further comprises a fifth indication of the first transmission direction, wherein, to select the first section of the surface of the wireless device for the first transmission direction, the at least one processor is configured to: select the first section of the surface of the wireless device for the first transmission direction based on the fifth indication of at least one of the transmissive signal configuration or the transmissive and reflective configuration.

8. The apparatus of claim 1, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration comprises a fourth indication of at least one of an incident angle or an incident distance for the first section of the surface of the wireless device for the first transmission direction and a fifth indication of at least one of a transmissive angle or a transmissive distance for the second section of the surface of the wireless device, wherein the at least one processor is further configured to: configure a first transmissive coefficient of a first set of meta-elements of the first section of the surface of the wireless device based on the fourth indication of at least one of the incident angle or the incident distance; and configure a second transmissive coefficient of a second set of meta-elements of the second section of the surface of the wireless device based on the fifth indication of at least one of the transmissive angle or the transmissive distance.

9. The apparatus of claim 8, wherein the incident angle is equal to the transmissive angle.

10. The apparatus of claim 8, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration further comprises a sixth indication of at least one of a minimum transmissive angle or a maximum transmissive angle and a seventh indication of at least one of a minimum incident angle or a maximum incident angle, wherein the at least one processor is further configured to: configure the first transmissive coefficient of the first set of meta-elements of the first section of the surface of the wireless device to perform beam sweeping based on at least one of the minimum transmissive angle or the maximum transmissive angle; and configure the second transmissive coefficient of the second set of meta-elements of the second section of the surface of the wireless device to perform the beam sweeping based on at least one of the minimum incident angle or the maximum incident angle.

11. The apparatus of claim 1, wherein at least one of the transmissive signal configuration or the transmissive and reflective configuration comprises a first beamforming gain factor, wherein the at

least one processor is further configured to: apply the first beamforming gain factor to at least one meta-element of the surface of the wireless device.

12. The apparatus of claim 11, wherein the at least one processor is further configured to: receive, from the network node, a second transmissive signal configuration or a second transmissive and reflective configuration comprising a second beamforming gain factor; and apply at least one of the second beamforming gain factor or the second beamforming gain factor and the first beamforming gain factor to the at least one meta-element of the surface of the wireless device.

13. The apparatus of claim 1, wherein the third indication of the transmissive and reflective capability indicates that a power split between transmission and reflection associated with a set of meta-elements of the surface of the wireless device is fixed.

14. The apparatus of claim 1, wherein the third indication of the transmissive and reflective capability indicates that a power split between transmission and reflection associated with meta-elements of the surface of the wireless device is configurable, wherein the transmissive and reflective configuration comprises a fourth indication of a power split ratio between the transmission and the reflection, wherein the at least one processor is further configured to: configure a set of meta-elements of the surface of the wireless device based on the fourth indication of the power split ratio between the transmission and the reflection.

15. The apparatus of claim 1, wherein the third section and the fourth section comprise an overlapping section of the surface of the wireless device, wherein, to select the third section of the surface of the wireless device for the transmissive mode, the at least one processor is configured to select a first set of time periods for the overlapping section of the surface of the wireless device, wherein, to select the fourth section of the surface of the wireless device for the reflective mode, the at least one processor is configured to select a second set of time periods for the overlapping section of the surface of the wireless device, wherein the first set of time periods and the second set of time periods are non-overlapping.

16. The apparatus of claim 1, wherein the at least one processor is further configured to: configure at least one of a transmissive coefficient for the third section of the surface or a reflective coefficient for the fourth section of the surface based on the transmissive and reflective configuration.

17. The apparatus of claim 1, further comprising a transceiver coupled to the at least one processor, wherein the at least one processor is configured to transmit the capability report via the transceiver, wherein the wireless device comprises a reconfigurable intelligent surface (RIS).

18. An apparatus for wireless communication at a network node, comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: receive, from a wireless device, a capability report comprising at least one of: (1) a first indication of a transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability; and transmit, for the wireless device based on the capability report, at least one of a transmissive signal configuration or a transmissive and reflective configuration associated with at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

19. The apparatus of claim 18, wherein the capability report comprises a fourth indication of a bidirectional transmission capability, wherein the at least one processor is further configured to: configure at least one of the transmissive signal configuration or the transmissive and reflective configuration to overlap the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is capable of simultaneous bidirectional transmission; and configure at least one of the transmissive signal configuration or the transmissive

and reflective configuration to split the surface of the wireless device into the first section for the first transmission direction and the second section for the second transmission direction based on the fourth indication of the bidirectional transmission capability indicating that the wireless device is not capable of the simultaneous bidirectional transmission.

20. (canceled)

21. (canceled)

22. (canceled)

23. (canceled)

24. (canceled)

25. (canceled)

26. (canceled)

27. (canceled)

28. (canceled)

29. A method of wireless communication at a wireless device, comprising: transmitting, for a network node, a capability report comprising at least one of: (1) a first indication of a first transmissive coefficient value, (2) a second indication of a sensing capability, or (3) a third indication of a transmissive and reflective capability; receiving, from the network node, at least one of a transmissive signal configuration or a transmissive and reflective configuration based on the capability report; and selecting, based on at least one of the transmissive signal configuration or the transmissive and reflective configuration, at least one of: (1) a first section of a surface of the wireless device for a first transmission direction, (2) a second section of the surface of the wireless device for a second transmission direction, (3) a third section of the surface of the wireless device for a transmissive mode, or (4) a fourth section of the surface of the wireless device for a reflective mode.

30. (canceled)
